

APPENDIX

Contents

A Useful Lemmas	13
A.1 General lemmas	13
A.2 Deterministic equivalents	14
A.3 Resolvent identities	16
B Random Matrix Analysis of the general model	19
B.1 Test expectation:	19
B.2 Test variance:	20
C Particular case: Isotropic covariance matrix	25
C.1 Resolvent identities in the case of $\mathbf{C} = \sigma^2 \mathbf{I}_p$	25
C.2 Test Expectation	26
C.3 Test Variance	26
D Random matrix analysis of distribution shift	29
D.1 Deterministic equivalents:	29
D.2 Deterministic equivalent of $\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}}$:	30
D.3 Useful results:	31
D.4 Test Expectation:	31
D.5 Test variance:	32
E Details about experiments with Safety LLM Alignment with IPO	34
E.1 Hyperparameters	34
F Details about experiments with LLM QA classification	35
F.1 Prompting LLMs	35
F.2 Hyper-parameters	35

A USEFUL LEMMAS

Notation: For $a \in \{1, 2\}$, we denote by $\mathbb{I}_a = \{i \mid \mathbf{x}_i \in \mathcal{C}_a\}$, i.e, the set of indices of vectors belonging to class $\mathcal{C}_a$. Furthermore, we denote $\Sigma = \boldsymbol{\mu}\boldsymbol{\mu}^\top + \mathbf{I}_p = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$ for $\mathbf{x} \in \mathcal{C}_a$, and $\Sigma_\beta = \boldsymbol{\mu}_\beta\boldsymbol{\mu}_\beta^\top + \mathbf{C}$

Here we will list the most useful lemmas and results used in our analysis.

A.1 GENERAL LEMMAS

Lemma A.1 (Inverse identity). *For invertible matrices $\mathbf{A}$ and $\mathbf{B}$, we have that:*

$$\mathbf{A}^{-1} - \mathbf{B}^{-1} = \mathbf{A}^{-1}(\mathbf{B} - \mathbf{A})\mathbf{B}^{-1}$$

Lemma A.2 (Woodbury). *For $\mathbf{A} \in \mathbb{R}^{p \times p}$, $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{p \times k}$, such that both $\mathbf{A}$ and $\mathbf{A} + \mathbf{U}\mathbf{V}^\top$ are invertible, we have:*

$$(\mathbf{A} + \mathbf{U}\mathbf{V}^\top)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{I}_k + \mathbf{V}^\top\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}^\top\mathbf{A}^{-1}$$

A particular case of this lemma A.2, in the case of $k = 1$, is called *Sherman-Morrisson's identity*.

Lemma A.3 (Sherman-Morrisson). *For $\mathbf{A} \in \mathbb{R}^{p \times p}$ invertible and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^p$, $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$ is invertible if and only if: $1 + \mathbf{v}^\top\mathbf{A}\mathbf{u} \neq 0$, and:*

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\top\mathbf{A}^{-1}}{1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}}$$

Besides,

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\top)^{-1}\mathbf{u} = \frac{\mathbf{A}^{-1}\mathbf{u}}{1 + \mathbf{v}^\top\mathbf{A}^{-1}\mathbf{u}}$$

A.2 DETERMINISTIC EQUIVALENTS

Let us state here the deterministic equivalent of the resolvent matrix $\mathbf{Q}$ defined in the general model's equation (6) for any general covariance matrix $\mathbf{C}$ and mean $\boldsymbol{\mu}_\beta = \beta\boldsymbol{\mu} + \boldsymbol{\mu}^\perp$ that define the statistic of the synthetic data, as in equation 13.

Lemma A.4 (Deterministic equivalent of $\mathbf{Q}$). *Under the 4.1 assumptions listed above in the main paper, a deterministic equivalent for $\mathbf{Q} \equiv \mathbf{Q}(\gamma)$, denoted $\bar{\mathbf{Q}}$, is given by:*

$$\bar{\mathbf{Q}} = \left(\frac{\pi(\boldsymbol{\mu}\boldsymbol{\mu}^\top + \mathbf{I}_p)}{1 + \delta} + \frac{\alpha(1 - \pi)(\boldsymbol{\mu}_\beta\boldsymbol{\mu}_\beta^\top + \mathbf{C})}{1 + \delta_S} + \gamma\mathbf{I}_p \right)^{-1}$$

where:

$$\pi = \frac{n}{n + m}, \quad \alpha = \phi(1 - \varepsilon) + \rho\varepsilon, \quad \delta = \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}), \quad \delta_S = \frac{\alpha}{N} \text{Tr}(\mathbf{C}\bar{\mathbf{Q}})$$

Proof. We want to find $\bar{\mathbf{Q}}$ such that for all bounded $\mathbf{a}, \mathbf{b} \in \mathbb{R}^p$:

$$\mathbf{a}^\top (\mathbb{E}[\mathbf{Q}] - \bar{\mathbf{Q}}) \mathbf{b} \rightarrow 0$$

Let $\bar{\mathbf{Q}} = (\mathbf{S} + \gamma\mathbf{I}_p)^{-1}$. We want to determine an $\mathbf{S}$ that satisfies the above property. We have that:

$$\begin{aligned} \mathbb{E}[\mathbf{Q}] - \bar{\mathbf{Q}} &= \mathbb{E}[\mathbf{Q}(\mathbf{S} - \frac{1}{N}\mathbf{V}\mathbf{V}^\top)\bar{\mathbf{Q}}] \quad (\text{lemma A.1}) \\ &= \frac{1}{N} \sum_{i=1}^N \mathbb{E}[\mathbf{Q}(\mathbf{S} - \mathbf{v}_i\mathbf{v}_i^\top)\bar{\mathbf{Q}}] \\ &= \frac{1}{N} \sum_{i=1}^n \mathbb{E}[\mathbf{Q}(\mathbf{S} - \mathbf{x}_i\mathbf{x}_i^\top)\bar{\mathbf{Q}}] + \frac{1}{N} \sum_{i=1}^m \mathbb{E}[\mathbf{Q}(\mathbf{S} - q_i\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top)\bar{\mathbf{Q}}] \\ &= \frac{1}{N} \sum_{i=1}^n \mathbb{E}[\mathbf{Q}\mathbf{S} - \frac{1}{1 + \delta_R}\mathbf{Q}_{-\mathbf{x}_i}\mathbf{x}_i\mathbf{x}_i^\top]\bar{\mathbf{Q}} + \frac{1}{N} \sum_{i=1}^m \mathbb{E}[\mathbf{Q}\mathbf{S} - \frac{q_i}{1 + \delta_S}\mathbf{Q}_{-\tilde{\mathbf{x}}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top]\bar{\mathbf{Q}} \\ &= \pi\mathbb{E}[\mathbf{Q}\mathbf{S} - \frac{1}{1 + \delta_R}\mathbf{Q}_{-\mathbf{x}_i}\mathbf{x}_i\mathbf{x}_i^\top]\bar{\mathbf{Q}} + (1 - \pi)\mathbb{E}[\mathbf{Q}\mathbf{S} - \frac{q_i}{1 + \delta_S}\mathbf{Q}_{-\tilde{\mathbf{x}}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top]\bar{\mathbf{Q}} \\ &= \pi\mathbb{E}[\mathbf{Q}_{-\mathbf{x}_i}(\mathbf{S} - \frac{\mathbf{x}_i\mathbf{x}_i^\top}{1 + \delta_R})\bar{\mathbf{Q}}] + (1 - \pi)\mathbb{E}[\mathbf{Q}_{-\tilde{\mathbf{x}}_i}(\mathbf{S} - \frac{q_i\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top}{1 + \delta_S})\bar{\mathbf{Q}}] + \mathcal{O}(N^{-1}) \end{aligned}$$

Thus, it suffices to have:

$$\mathbf{S} = \frac{\pi(\boldsymbol{\mu}\boldsymbol{\mu}^\top + \mathbf{I}_p)}{1 + \delta_R} + \frac{\alpha(1 - \pi)(\boldsymbol{\mu}_\beta\boldsymbol{\mu}_\beta^\top + \mathbf{C})}{1 + \delta_S}$$

to get the desired property. $\square$

Lemma A.5 (Deterministic equivalent of $\mathbf{Q}\mathbf{A}\mathbf{Q}$). *Let $\mathbf{A} \in \mathbb{R}^{p \times p}$ be any deterministic symmetric semi-definite matrix. We have that:*

$$\mathbf{Q}\mathbf{A}\mathbf{Q} \leftrightarrow \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} + \frac{\pi}{N(1 + \delta)^2} \text{Tr}(\Sigma\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}] + \frac{\alpha(1 - \pi)}{N(1 + \delta_S)^2} \text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma_\beta\mathbf{Q}]$$

Thus, we get that for $\mathbf{A} = \Sigma$, and for $\mathbf{A} = \Sigma_\beta$:

$$\begin{aligned} \mathbf{Q}\Sigma\mathbf{Q} &\leftrightarrow \bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}} + \frac{\pi}{N(1 + \delta)^2} \text{Tr}((\Sigma\bar{\mathbf{Q}})^2)\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}] + \frac{\alpha(1 - \pi)}{N(1 + \delta_S)^2} \text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma_\beta\mathbf{Q}] \\ \mathbf{Q}\Sigma_\beta\mathbf{Q} &\leftrightarrow \bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}} + \frac{\pi}{N(1 + \delta)^2} \text{Tr}(\Sigma\bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}] + \frac{\alpha(1 - \pi)}{N(1 + \delta_S)^2} \text{Tr}((\Sigma_\beta\bar{\mathbf{Q}})^2)\mathbb{E}[\mathbf{Q}\Sigma_\beta\mathbf{Q}] \end{aligned}$$

And by denoting:

$$\begin{aligned} a_1 &= \frac{\pi}{N(1 + \delta)^2} \text{Tr}((\Sigma\bar{\mathbf{Q}})^2), \quad b_1 = \frac{\alpha(1 - \pi)}{N(1 + \delta_S)^2} \text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}}), \\ a_2 &= \frac{\pi}{N(1 + \delta)^2} \text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}}), \quad b_2 = \frac{\alpha(1 - \pi)}{N(1 + \delta_S)^2} \text{Tr}((\Sigma_\beta\bar{\mathbf{Q}})^2) \\ h &= (1 - b_2)(1 - a_1) - a_2b_1 \end{aligned}$$

We get that:

$$\begin{aligned}\mathbf{Q}\Sigma\mathbf{Q} &\leftrightarrow \frac{1-b_2}{h}\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}} + \frac{b_1}{h}\bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}}, \\ \mathbf{Q}\Sigma_\beta\mathbf{Q} &\leftrightarrow \frac{a_2}{h}\bar{\mathbf{Q}}\Sigma\bar{\mathbf{Q}} + \frac{1-a_1}{h}\bar{\mathbf{Q}}\Sigma_\beta\bar{\mathbf{Q}}.\end{aligned}$$

Proof. Recall that:

$$\bar{\mathbf{Q}}(\gamma) = \left(\frac{\pi\Sigma}{1+\delta} + \frac{\alpha(1-\pi)\Sigma_\beta}{1+\delta_S} + \gamma\mathbf{I}_p \right)^{-1}$$

Let us denote by : $\mathbf{S} = \frac{\pi\Sigma}{1+\delta} + \frac{\alpha(1-\pi)\Sigma_\beta}{1+\delta_S}$, so that: $\bar{\mathbf{Q}} = (\mathbf{S} + \gamma\mathbf{I}_p)^{-1}$.

We have that:

$$\begin{aligned}\mathbb{E}[\mathbf{Q}\mathbf{A}\mathbf{Q}] &= \mathbb{E}[\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] + \mathbb{E}[(\mathbf{Q} - \bar{\mathbf{Q}})\mathbf{A}\mathbf{Q}] \\ &= \bar{\mathbf{Q}}\mathbb{E}[\mathbf{A}\mathbf{Q}] + \mathbb{E}[(\mathbf{Q} - \bar{\mathbf{Q}})\mathbf{A}\mathbf{Q}] \\ &= \bar{\mathbf{Q}}(\mathbb{E}[\mathbf{A}\bar{\mathbf{Q}}] + \mathbb{E}[\mathbf{A}(\mathbf{Q} - \bar{\mathbf{Q}})]) + \mathbb{E}[(\mathbf{Q} - \bar{\mathbf{Q}})\mathbf{A}\mathbf{Q}] \\ &= \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} + \mathbb{E}[(\mathbf{Q} - \bar{\mathbf{Q}})\mathbf{A}\mathbf{Q}] \\ &= \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} + \mathbb{E}\left[\mathbf{Q}\left(\mathbf{S} - \frac{1}{N}\mathbf{V}\mathbf{V}^\top\right)\mathbf{A}\mathbf{Q}\right] \\ &= \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} + \mathbb{E}[\mathbf{Q}\mathbf{S}\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \frac{1}{N}\sum_{i=1}^N\mathbb{E}[\mathbf{Q}\mathbf{v}_i\mathbf{v}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] \\ &= \bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}} + \mathbb{E}[\mathbf{Q}\mathbf{S}\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \pi\mathbb{E}[\mathbf{Q}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - (1-\pi)\mathbb{E}[\mathbf{Q}q_i\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}]\end{aligned}$$

And we have that:

$$\begin{aligned}\mathbb{E}[\mathbf{Q}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] &= \frac{1}{1+\delta}\mathbb{E}[\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] \\ &= \frac{1}{1+\delta}\mathbb{E}\left[\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\left(\mathbf{Q}_{-x_i} - \frac{\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\mathbf{Q}_{-x_i}}{N(1+\delta)}\right)\right] \\ &= \frac{1}{1+\delta}\mathbb{E}[\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}_{-x_i}] - \frac{1}{N(1+\delta)^2}\mathbb{E}[\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\mathbf{Q}_{-x_i}] \\ &= \frac{1}{1+\delta}\mathbb{E}[\mathbf{Q}\Sigma\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \frac{1}{N(1+\delta)^2}\text{Tr}(\Sigma\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}_{-x_i}\mathbf{x}_i\mathbf{x}_i^\top\mathbf{Q}_{-x_i}] \\ &= \frac{1}{1+\delta}\mathbb{E}[\mathbf{Q}\Sigma\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \frac{1}{N(1+\delta)^2}\text{Tr}(\Sigma\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]\end{aligned}$$

And:

$$\begin{aligned}\mathbb{E}[q_i\mathbf{Q}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] &= \frac{1}{1+\delta_S}\mathbb{E}[q_i\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] \\ &= \frac{1}{1+\delta_S}\mathbb{E}\left[q_i\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\left(\mathbf{Q}_{-\tilde{x}_i} - \frac{q_i\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\mathbf{Q}_{-\tilde{x}_i}}{N(1+\delta_S)}\right)\right] \\ &= \frac{1}{1+\delta_S}\mathbb{E}[q_i\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}_{-\tilde{x}_i}] - \frac{1}{N(1+\delta_S)^2}\mathbb{E}[q_i\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\mathbf{Q}_{-\tilde{x}_i}] \\ &= \frac{\alpha}{1+\delta_S}\mathbb{E}[\mathbf{Q}\Sigma_\beta\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \frac{\alpha}{N(1+\delta_S)^2}\text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}_{-\tilde{x}_i}\tilde{\mathbf{x}}_i\tilde{\mathbf{x}}_i^\top\mathbf{Q}_{-\tilde{x}_i}] \\ &= \frac{\alpha}{1+\delta_S}\mathbb{E}[\mathbf{Q}\Sigma_\beta\bar{\mathbf{Q}}\mathbf{A}\mathbf{Q}] - \frac{\alpha}{N(1+\delta_S)^2}\text{Tr}(\Sigma_\beta\bar{\mathbf{Q}}\mathbf{A}\bar{\mathbf{Q}})\mathbb{E}[\mathbf{Q}\Sigma_\beta\mathbf{Q}]\end{aligned}$$

Which concludes the proof by summing all these separate terms. $\square$

Corollary A.6 (Trace identities). *Using the above lemma A.5, we get that:*

$$\frac{1}{N}\text{Tr}(\Sigma\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]) = \frac{(1+\delta)^2}{\pi h}(a_1(1-b_2) + a_2b_1), \quad \frac{1}{N}\text{Tr}(\Sigma_\beta\mathbb{E}[\mathbf{Q}\Sigma\mathbf{Q}]) = \frac{(1+\delta_S)^2}{\alpha(1-\pi)h}b_1$$